

**KARGOPOL', Russian Federation**WMO: **228450**Lat: **61.5067N**Lon: **38.9347E**Elev: **126**StdP: **99.82**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -30.1      | -26.7 | -32.6                         | 0.2 | -30.1 | -29.2 | 0.3 | -26.8 | 7.3                   | -5.0 | 6.5 | -4.6 | 0.9                   | 20   | 0.603 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.1                    | 27.5            | 19.5 | 25.6 | 18.6 | 23.7 | 17.5 | 20.9                | 25.7 | 19.7 | 24.0 | 18.5 | 22.5 | 2.5                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.2                            | 14.2 | 23.4 | 18.0 | 13.2 | 21.8 | 16.9 | 12.2 | 21.0 | 61.0          | 26.0 | 56.8 | 24.1 | 53.1 | 22.5 | 26.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 6.4               | 5.7  | 5.2 | -32.1                      | 29.9 | 4.1                | 2.1 | -35.0                                              | 31.3 | -37.4      | 32.6 | -39.7      | 33.7 | -42.7      | 35.2 |
|                   |      |     | -32.2                      | 22.4 | 4.1                | 1.8 | -35.1                                              | 23.7 | -37.5      | 24.7 | -39.8      | 25.7 | -42.8      | 27.0 |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |     |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.8    | -11.1 | -10.1 | -4.9  | 2.4   | 9.2   | 14.1  | 17.4  | 14.6  | 9.6   | 3.0   | -3.6  | -7.6 |
|                                                                                  | DBStd       | 11.21  | 8.31  | 7.82  | 5.24  | 4.68  | 5.15  | 4.32  | 3.80  | 3.63  | 3.78  | 4.08  | 6.03  | 7.33 |
|                                                                                  | HDD10.0     | 3234   | 655   | 563   | 462   | 233   | 79    | 10    | 4     | 53    | 218   | 409   | 546   |      |
|                                                                                  | HDD18.3     | 5722   | 913   | 797   | 721   | 478   | 286   | 139   | 63    | 124   | 263   | 474   | 659   | 804  |
|                                                                                  | CDD10.0     | 613    | 0     | 0     | 0     | 4     | 53    | 134   | 231   | 148   | 39    | 2     | 0     | 0    |
|                                                                                  | CDD18.3     | 59     | 0     | 0     | 0     | 0     | 2     | 13    | 34    | 9     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 483    | 0     | 0     | 0     | 1     | 24    | 127   | 258   | 72    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 92     | 0     | 0     | 0     | 0     | 2     | 27    | 51    | 12    | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.4    | 2.5   | 2.6   | 2.6   | 2.3   | 2.4   | 2.2   | 2.0   | 2.0   | 2.2   | 2.6   | 2.7   | 2.8  |
| Precipitation                                                                    | PrecAvg     | 616    | 34    | 26    | 31    | 39    | 51    | 70    | 71    | 83    | 61    | 60    | 47    | 43   |
|                                                                                  | PrecMax     | 797    | 60    | 54    | 67    | 88    | 124   | 134   | 139   | 255   | 112   | 117   | 88    | 78   |
|                                                                                  | PrecMin     | 417    | 8     | 3     | 2     | 5     | 16    | 6     | 22    | 20    | 13    | 2     | 11    | 10   |
|                                                                                  | PrecStd     | 82     | 10    | 13    | 18    | 21    | 25    | 33    | 29    | 48    | 25    | 25    | 18    | 17   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 1.9   | 2.6   | 8.0   | 20.0  | 26.5  | 29.8  | 30.5  | 28.2  | 21.8  | 14.8  | 7.3   | 4.8  |
|                                                                                  |             | MCWB   | 1.0   | 1.6   | 3.0   | 11.7  | 16.1  | 20.3  | 21.6  | 20.8  | 16.7  | 11.6  | 6.4   | 4.2  |
|                                                                                  | 2%          | DB     | 1.0   | 1.4   | 4.9   | 15.6  | 23.1  | 26.7  | 28.0  | 25.2  | 19.0  | 11.8  | 5.7   | 2.6  |
|                                                                                  |             | MCWB   | 0.4   | 0.7   | 1.5   | 9.1   | 14.3  | 18.4  | 20.2  | 18.6  | 15.3  | 10.2  | 4.9   | 2.1  |
|                                                                                  | 5%          | DB     | 0.4   | 0.6   | 3.1   | 12.0  | 20.4  | 24.4  | 26.2  | 22.9  | 16.9  | 10.0  | 4.4   | 1.4  |
|                                                                                  |             | MCWB   | -0.1  | 0.0   | 0.9   | 6.8   | 13.1  | 17.4  | 19.6  | 17.6  | 13.7  | 9.0   | 3.8   | 0.9  |
|                                                                                  | 10%         | DB     | -0.5  | -0.3  | 1.8   | 9.5   | 18.1  | 22.1  | 24.4  | 21.0  | 15.2  | 8.5   | 3.0   | 0.6  |
|                                                                                  |             | MCWB   | -0.9  | -0.8  | 0.3   | 5.3   | 11.8  | 16.0  | 18.6  | 16.5  | 12.9  | 7.6   | 2.6   | 0.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 1.2   | 1.8   | 3.7   | 12.4  | 18.0  | 21.6  | 22.9  | 21.8  | 17.4  | 12.1  | 6.8   | 4.3  |
|                                                                                  |             | MCDB   | 1.5   | 2.4   | 6.6   | 20.0  | 23.6  | 27.6  | 28.1  | 26.5  | 20.6  | 14.0  | 7.0   | 4.6  |
|                                                                                  | 2%          | WB     | 0.6   | 0.8   | 2.3   | 9.5   | 15.9  | 19.9  | 21.4  | 19.5  | 15.7  | 10.3  | 5.0   | 2.2  |
|                                                                                  |             | MCDB   | 0.9   | 1.2   | 4.1   | 14.4  | 21.1  | 25.3  | 26.4  | 23.5  | 18.2  | 11.3  | 5.5   | 2.6  |
|                                                                                  | 5%          | WB     | 0.0   | 0.1   | 1.3   | 7.6   | 14.1  | 18.3  | 20.3  | 18.2  | 14.3  | 9.1   | 3.8   | 1.0  |
|                                                                                  |             | MCDB   | 0.4   | 0.6   | 2.5   | 11.3  | 18.9  | 23.1  | 24.7  | 21.6  | 16.3  | 10.1  | 4.3   | 1.3  |
|                                                                                  | 10%         | WB     | -1.0  | -0.8  | 0.6   | 5.7   | 12.5  | 16.7  | 19.3  | 17.0  | 13.2  | 7.6   | 2.6   | 0.2  |
|                                                                                  |             | MCDB   | -0.5  | -0.2  | 1.7   | 9.1   | 17.4  | 20.8  | 23.3  | 20.3  | 14.8  | 8.4   | 2.9   | 0.6  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.9   | 6.8   | 8.2   | 9.0   | 9.8   | 9.4   | 9.1   | 8.4   | 6.8   | 4.2   | 4.0   | 4.8  |
|                                                                                  |             | MCDBR  | 4.6   | 4.4   | 8.6   | 13.2  | 13.5  | 12.6  | 11.6  | 11.2  | 9.6   | 6.0   | 4.0   | 3.2  |
|                                                                                  |             | MCWBR  | 4.6   | 4.3   | 6.2   | 7.8   | 6.4   | 5.9   | 5.3   | 5.4   | 5.7   | 4.6   | 3.9   | 3.1  |
|                                                                                  | 5% WB       | MCDBR  | 4.8   | 4.5   | 6.4   | 11.9  | 11.5  | 11.1  | 10.3  | 9.5   | 7.9   | 5.8   | 3.8   | 3.2  |
|                                                                                  |             | MCWBR  | 4.8   | 4.4   | 4.8   | 7.5   | 6.2   | 5.8   | 5.4   | 5.2   | 5.2   | 4.7   | 3.7   | 3.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.247  | 0.295 | 0.321 | 0.361 | 0.352 | 0.351 | 0.386 | 0.369 | 0.337 | 0.318 | 0.269 | 0.216 |      |
|                                                                                  | taud        | 2.023  | 2.077 | 2.120 | 2.241 | 2.443 | 2.472 | 2.386 | 2.412 | 2.488 | 2.389 | 2.127 | 1.956 |      |
|                                                                                  | Ebn at Noon | 441    | 650   | 778   | 824   | 871   | 877   | 830   | 814   | 780   | 649   | 414   | 296   |      |
|                                                                                  | Edh at Noon | 40     | 75    | 102   | 111   | 99    | 98    | 105   | 94    | 73    | 56    | 36    | 24    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.25   | 0.82  | 2.08  | 3.58  | 4.93  | 5.27  | 5.15  | 3.76  | 2.14  | 0.85  | 0.24  | 0.12  |      |
|                                                                                  | RadStd      | 0.03   | 0.20  | 0.30  | 0.38  | 0.37  | 0.43  | 0.54  | 0.34  | 0.22  | 0.10  | 0.04  | 0.02  |      |

**Historical Trends**

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only           | +0.68 | N/A     | N/A    | N/A     | N/A   | N/A   | -216        | -254    | N/A     | N/A     |
| (47) Regional (6 neighbors) | +0.64 | N/A     | N/A    | N/A     | N/A   | N/A   | -198        | -217    | N/A     | N/A     |

Nomenclature: See separate page